

## Recall Quotient

step 1 Equivalence classes  $\Leftrightarrow$  partition

step 2 Define operation on Equivalence classes  $S_i \odot S_j = S_{kl} (i, j)$

Ex.  $V/U, U \in \mathcal{L}(V) \Rightarrow V = \dot{\bigcup}_{u \in V/U} (u + U)$

$(V/U, +, \cdot)$  is a quotient space of  $V$  by  $U$

Ex  $\mathbb{C} \cong \mathbb{R}[x] / (x^2 + 1) \mathbb{R}[x]$

Ex  $\mathbb{C} \cong \mathbb{R}$

Thm Let  $\dim V = n < +\infty$ , Then  $\dim V/U = \dim V - \dim U$

In particular, if  $\alpha_1, \dots, \alpha_s$  is a basis of  $U$ ,  $\alpha_1, \alpha_2, \dots, \alpha_s, \alpha_{s+1}, \dots, \alpha_n$  a basis

of  $V$ , Then  $\{\alpha_{s+1} + U, \dots, \alpha_n + U\}$  is a basis of  $V/U$

~~Isomorphism Thm~~ Let  $a, b \in \mathcal{L}(V)$  Then

1)  $a+b/a \cong b/anb$

(projection

2)  $a/anb \cong a+b/b$

property)

3)  $a+b/anb = a/anb \oplus b/anb$  (in  $\mathcal{L}(V/anb)$ )



New  $V, U \subseteq L(V) \Rightarrow V/U$

$\text{Hom}(V, V/U)$

Then  $0 \rightarrow V \xrightarrow{\pi} V/U$  quotient map

$$v \mapsto v + U$$

$$\text{Im } \pi = V/U, \text{ Ker } \pi = U$$

Given  $V, W/F, \sigma \in \text{Hom}(V, W)$

$$\text{Ker } \sigma \xrightarrow{i} V \xrightarrow{\sigma} W$$

$$\downarrow \begin{matrix} \bar{i} \\ \text{Im } \sigma \end{matrix} \nearrow$$

$\bar{\sigma} \in \text{Hom}(V, \text{Im } \sigma)$   $\bar{\sigma}$  is induced by  $\sigma$

$$\text{Ker } \sigma \xrightarrow{i} V \xrightarrow{\bar{\sigma}} \text{Im } \sigma$$

Get a new one  $\bar{\sigma} \in \text{Hom}(\text{Ker } \sigma, \text{Im } \sigma)$

$$v_1 \xrightarrow{f_1} v_2 \xrightarrow{f_2} v_3 \dots \xrightarrow{f_k} v_k \rightarrow \dots$$

$$f_{i+1} \circ f_i = 0, \forall i \text{ (complex, 复形)}$$

$$\text{Im } f_i \subseteq \text{Ker } f_{i+1} \Rightarrow \text{Ker } f_{i+1} / \text{Im } f_i = H_i(V)$$

Homology (同调)

• Computation: known  $\text{Im } i \subseteq \text{Ker } \bar{\sigma}$

$$\text{so } 0 = \text{Ker } \sigma / \text{Im } i = \text{Ker } \bar{\sigma} / \text{Im } i$$

$$v_1 \xrightarrow{f_1} v_2 \xrightarrow{f_2} v_3 \quad \text{Im } f_1 = \text{Ker } f_2 \quad [\text{Exact at } v_2 \text{ (在 } v_2 \text{ 处匹配)}]$$

• Special Example

$$\text{Ker } f_1 = 0 \rightarrow v_1 \xrightarrow{f_1} v_2 \xrightarrow{f_2} v_3 \rightarrow 0 \quad [\text{short Exact sequence}]$$

$$\downarrow \text{Im } f_1 = \text{Ker } f_2$$

$$\text{Conclusion: } \text{Ker } \sigma \xrightarrow{i} V \xrightarrow{\sigma} W \xrightarrow{\pi} W / \text{Im } \sigma$$

$$V / \text{Ker } \sigma \cong \text{Im } \sigma$$

## Dual Space

Given  $V$ , First compare  $\text{Hom}(V, \mathbb{F})$

$$\sigma: V \longrightarrow \mathbb{F}$$

$$v \longmapsto \sigma(v)$$

Limit your study on easy  $V = \mathbb{F}^n \rightarrow (x_1, \dots, x_n) = x \quad r(v) = \sum_{i=1}^n a_i x_i = (a_1, \dots, a_n)x$

Generally,  $\dim V = n < \infty$ , Fix a basis  $v_1, \dots, v_n$  of  $V$

Then  $\sigma(v) = \sum_{i=1}^n a_i x_i$  where

$$\sigma(v_1, v_2, \dots, v_n)x = \sigma\left(\sum_{i=1}^n x_i v_i\right) = \sum_{i=1}^n \sigma(x_i v_i)$$

or Let  $\dim W = m$ , Then  $\text{Hom}(V, W) \cong \text{Hom}(V, \mathbb{F}^m) \cong \text{Hom}(V, \mathbb{F})^{\oplus m}$ .

RK. key object  $\text{Hom}(V, \mathbb{F}) \cong V^*$  (dual space of  $V$  / conjugate space)

Ex.  $\text{Hom}(\mathbb{F}[x], \mathbb{F}) = \mathbb{F}[x]^* \cong \mathbb{F}[x]$

so  $V \not\cong V^*$  but  $V \cong V^*$  if  $V$  is f.d.

Thm There is a distinguished map  $V \xrightarrow{E} V^{**}$  defined by

$$v \xrightarrow{E} V^{**} = \text{Hom}(V^*, \mathbb{F}) = \text{Hom}(\text{Hom}(V, \mathbb{F}), \mathbb{F})$$

$$v \longmapsto E(v) : f \mapsto E(v)(f) = f(v)$$

evaluation map (对值映射)

Moreover,  $\ker(Ev) = 0$

Pf. Let  $v \in \ker(Ev)$  Then  $Ev(v) = 0$ , so  $\forall f \in V^* f(v) = 0 \Rightarrow v = 0$

Let  $0 \neq v \in V$ , find  $f \in V^*$  s.t.  $f(v) \neq 0$

Define  $f(v) = 1$ ,  $f(Rv) = 0$

$\forall u \in V$   $u \notin \mathbb{F}v$ ,  $f(u) = 0$

or. If  $V$  is f.d. Then  $Ev$  is an isom.